

Part I - Prosper Loan Dataset

by S Sarveshwar

Preliminary Wrangling

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
%matplotlib inline
```

```
sns.set_style('whitegrid')
```

```
df = pd.read_csv('prosperLoanData.csv')
df.head()
```

	ListingKey	ListingNumber	ListingCreationDate \
0	1021339766868145413AB3B	193129	2007-08-26 19:09:29.263000000
1	10273602499503308B223C1	1209647	2014-02-27 08:28:07.900000000
2	0EE9337825851032864889A	81716	2007-01-05 15:00:47.090000000
3	0EF5356002482715299901A	658116	2012-10-22 11:02:35.010000000
4	0F023589499656230C5E3E2	909464	2013-09-14 18:38:39.097000000

	CreditGrade	Term	LoanStatus	ClosedDate	BorrowerAPR \
0	C	36	Completed	2009-08-14 00:00:00	0.16516
1	NaN	36	Current	NaN	0.12016
2	HR	36	Completed	2009-12-17 00:00:00	0.28269
3	NaN	36	Current	NaN	0.12528
4	NaN	36	Current	NaN	0.24614

	BorrowerRate	LenderYield	...	LP_ServiceFees
LP_CollectionFees \				
0	0.1580	0.1380	...	-133.18 0.0
1	0.0920	0.0820	...	0.00 0.0
2	0.2750	0.2400	...	-24.20 0.0
3	0.0974	0.0874	...	-108.01 0.0

```
4          0.2085          0.1985    ...          -60.27          0.0
```

```
LP_GrossPrincipalLoss LP_NetPrincipalLoss
```

```
LP_NonPrincipalRecoverypayments \
```

```
0          0.0          0.0
```

```
0.0
```

```
1          0.0          0.0
```

```
0.0
```

```
2          0.0          0.0
```

```
0.0
```

```
3          0.0          0.0
```

```
0.0
```

```
4          0.0          0.0
```

```
0.0
```

```
PercentFunded Recommendations InvestmentFromFriendsCount \
```

```
0          1.0          0          0
```

```
1          1.0          0          0
```

```
2          1.0          0          0
```

```
3          1.0          0          0
```

```
4          1.0          0          0
```

```
InvestmentFromFriendsAmount Investors
```

```
0          0.0          258
```

```
1          0.0          1
```

```
2          0.0          41
```

```
3          0.0          158
```

```
4          0.0          20
```

```
[5 rows x 81 columns]
```

```
df.shape
```

```
df.info()
```

```
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 113937 entries, 0 to 113936
```

```
Data columns (total 81 columns):
```

#	Column	Non-Null Count	Dtype
0	ListingKey	113937 non-null	object
1	ListingNumber	113937 non-null	int64
2	ListingCreationDate	113937 non-null	object
3	CreditGrade	28953 non-null	object
4	Term	113937 non-null	int64
5	LoanStatus	113937 non-null	object
6	ClosedDate	55089 non-null	object
7	BorrowerAPR	113912 non-null	float64
8	BorrowerRate	113937 non-null	float64

9	LenderYield	113937	non-null	float64
10	EstimatedEffectiveYield	84853	non-null	float64
11	EstimatedLoss	84853	non-null	float64
12	EstimatedReturn	84853	non-null	float64
13	ProsperRating (numeric)	84853	non-null	float64
14	ProsperRating (Alpha)	84853	non-null	object
15	ProsperScore	84853	non-null	float64
16	ListingCategory (numeric)	113937	non-null	int64
17	BorrowerState	108422	non-null	object
18	Occupation	110349	non-null	object
19	EmploymentStatus	111682	non-null	object
20	EmploymentStatusDuration	106312	non-null	float64
21	IsBorrowerHomeowner	113937	non-null	bool
22	CurrentlyInGroup	113937	non-null	bool
23	GroupKey	13341	non-null	object
24	DateCreditPulled	113937	non-null	object
25	CreditScoreRangeLower	113346	non-null	float64
26	CreditScoreRangeUpper	113346	non-null	float64
27	FirstRecordedCreditLine	113240	non-null	object
28	CurrentCreditLines	106333	non-null	float64
29	OpenCreditLines	106333	non-null	float64
30	TotalCreditLinespast7years	113240	non-null	float64
31	OpenRevolvingAccounts	113937	non-null	int64
32	OpenRevolvingMonthlyPayment	113937	non-null	float64
33	InquiriesLast6Months	113240	non-null	float64
34	TotalInquiries	112778	non-null	float64
35	CurrentDelinquencies	113240	non-null	float64
36	AmountDelinquent	106315	non-null	float64
37	DelinquenciesLast7Years	112947	non-null	float64
38	PublicRecordsLast10Years	113240	non-null	float64
39	PublicRecordsLast12Months	106333	non-null	float64
40	RevolvingCreditBalance	106333	non-null	float64
41	BankcardUtilization	106333	non-null	float64
42	AvailableBankcardCredit	106393	non-null	float64
43	TotalTrades	106393	non-null	float64
44	TradesNeverDelinquent (percentage)	106393	non-null	float64
45	TradesOpenedLast6Months	106393	non-null	float64
46	DebtToIncomeRatio	105383	non-null	float64
47	IncomeRange	113937	non-null	object
48	IncomeVerifiable	113937	non-null	bool
49	StatedMonthlyIncome	113937	non-null	float64
50	LoanKey	113937	non-null	object
51	TotalProsperLoans	22085	non-null	float64
52	TotalProsperPaymentsBilled	22085	non-null	float64
53	OnTimeProsperPayments	22085	non-null	float64
54	ProsperPaymentsLessThanOneMonthLate	22085	non-null	float64
55	ProsperPaymentsOneMonthPlusLate	22085	non-null	float64
56	ProsperPrincipalBorrowed	22085	non-null	float64
57	ProsperPrincipalOutstanding	22085	non-null	float64

58	ScorexChangeAtTimeOfListing	18928	non-null	float64
59	LoanCurrentDaysDelinquent	113937	non-null	int64
60	LoanFirstDefaultedCycleNumber	16952	non-null	float64
61	LoanMonthsSinceOrigination	113937	non-null	int64
62	LoanNumber	113937	non-null	int64
63	LoanOriginalAmount	113937	non-null	int64
64	LoanOriginationDate	113937	non-null	object
65	LoanOriginationQuarter	113937	non-null	object
66	MemberKey	113937	non-null	object
67	MonthlyLoanPayment	113937	non-null	float64
68	LP_CustomerPayments	113937	non-null	float64
69	LP_CustomerPrincipalPayments	113937	non-null	float64
70	LP_InterestandFees	113937	non-null	float64
71	LP_ServiceFees	113937	non-null	float64
72	LP_CollectionFees	113937	non-null	float64
73	LP_GrossPrincipalLoss	113937	non-null	float64
74	LP_NetPrincipalLoss	113937	non-null	float64
75	LP_NonPrincipalRecoverypayments	113937	non-null	float64
76	PercentFunded	113937	non-null	float64
77	Recommendations	113937	non-null	int64
78	InvestmentFromFriendsCount	113937	non-null	int64
79	InvestmentFromFriendsAmount	113937	non-null	float64
80	Investors	113937	non-null	int64

dtypes: bool(3), float64(50), int64(11), object(17)

memory usage: 68.1+ MB

	ListingNumber	Term	BorrowerAPR	BorrowerRate \
count	1.139370e+05	113937.000000	113912.000000	113937.000000
mean	6.278857e+05	40.830248	0.218828	0.192764
std	3.280762e+05	10.436212	0.080364	0.074818
min	4.000000e+00	12.000000	0.006530	0.000000
25%	4.009190e+05	36.000000	0.156290	0.134000
50%	6.005540e+05	36.000000	0.209760	0.184000
75%	8.926340e+05	36.000000	0.283810	0.250000
max	1.255725e+06	60.000000	0.512290	0.497500

	LenderYield	EstimatedEffectiveYield	EstimatedLoss	
EstimatedReturn \				
count	113937.000000	84853.000000	84853.000000	
mean	0.182701	0.168661	0.080306	
std	0.074516	0.068467	0.046764	
min	-0.010000	-0.182700	0.004900	-
25%	0.124200	0.115670	0.042400	
50%	0.173000	0.161500	0.072400	
max	0.091700			

75%	0.240000	0.224300	0.112000
0.116600			
max	0.492500	0.319900	0.366000
0.283700			
	ProsperRating (numeric)	ProsperScore	LP_ServiceFees \
count	84853.000000	84853.000000	113937.000000
mean	4.072243	5.950067	-54.725641
std	1.673227	2.376501	60.675425
min	1.000000	1.000000	-664.870000
25%	3.000000	4.000000	-73.180000
50%	4.000000	6.000000	-34.440000
75%	5.000000	8.000000	-13.920000
max	7.000000	11.000000	32.060000
	LP_CollectionFees	LP_GrossPrincipalLoss	
LP_NetPrincipalLoss \			
count	113937.000000	113937.000000	113937.000000
mean	-14.242698	700.446342	681.420499
std	109.232758	2388.513831	2357.167068
min	-9274.750000	-94.200000	-954.550000
25%	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000
max	0.000000	25000.000000	25000.000000
	LP_NonPrincipalRecoverypayments	PercentFunded	Recommendations
\			
count	113937.000000	113937.000000	113937.000000
mean	25.142686	0.998584	0.048027
std	275.657937	0.017919	0.332353
min	0.000000	0.700000	0.000000
25%	0.000000	1.000000	0.000000
50%	0.000000	1.000000	0.000000
75%	0.000000	1.000000	0.000000
max	21117.900000	1.012500	39.000000

	InvestmentFromFriendsCount	InvestmentFromFriendsAmount
Investors		
count	113937.000000	113937.000000
113937.000000		
mean	0.023460	16.550751
80.475228		
std	0.232412	294.545422
103.239020		
min	0.000000	0.000000
1.000000		
25%	0.000000	0.000000
2.000000		
50%	0.000000	0.000000
44.000000		
75%	0.000000	0.000000
115.000000		
max	33.000000	25000.000000
1189.000000		

[8 rows x 61 columns]

df.isnull().sum()

ListingKey	0
ListingNumber	0
ListingCreationDate	0
CreditGrade	84984
Term	0
...	
PercentFunded	0
Recommendations	0
InvestmentFromFriendsCount	0
InvestmentFromFriendsAmount	0
Investors	0

Length: 81, dtype: int64

df.sample(10)

	ListingKey	ListingNumber	
ListingCreationDate \			
35423	BC6234226312131416C8DA3	348093	2008-06-10
13:12:13.567000000			
7985	99ED34789489579407D3954	451567	2010-03-23
19:23:05.040000000			
86362	EEF2359117967217506FDDD	945349	2013-10-10
08:26:35.750000000			
78863	1B2B3544420119669E13F46	574933	2012-04-03
08:29:54.537000000			

48804	3B21357593798573840A41E	750980	2013-04-12
11:06:31.877000000			
8576	38223600968175199FF2648	1176567	2014-01-31
14:26:22.843000000			
34564	322C355516872300479F684	624794	2012-08-15
16:28:03.747000000			
112712	E2CD341127391966656EBA3	266982	2008-01-17
10:03:22.690000000			
82096	6C4F341157596208741FEEA	275695	2008-02-04
12:40:16.957000000			
90139	C3F13419756001014EAC3A6	319283	2008-04-27
15:20:12.330000000			

	CreditGrade	Term	LoanStatus	ClosedDate	BorrowerAPR
\					
35423	B	36	Completed	2010-04-08 00:00:00	0.15882
7985	NaN	36	Completed	2012-05-10 00:00:00	0.26282
86362	NaN	60	Current	NaN	0.28078
78863	NaN	12	Completed	2013-03-18 00:00:00	0.26584
48804	NaN	36	Current	NaN	0.29537
8576	NaN	36	Current	NaN	0.22773
34564	NaN	36	Current	NaN	0.19645
112712	D	36	Chargedoff	2009-09-24 00:00:00	0.25517
82096	A	36	Completed	2011-01-31 00:00:00	0.17445
90139	D	36	Completed	2011-05-09 00:00:00	0.17677

	BorrowerRate	LenderYield	...	LP_ServiceFees
LP_CollectionFees \				
35423	0.1445	0.1345	...	-178.91
0.0				
7985	0.2399	0.2299	...	-73.46
0.0				
86362	0.2549	0.2449	...	-16.92
0.0				
78863	0.1672	0.1572	...	-10.17
0.0				
48804	0.2566	0.2466	...	-33.61
0.0				
8576	0.1905	0.1805	...	0.00
0.0				

34564	0.1599	0.1499	...	-213.74
0.0				
112712	0.2400	0.2300	...	-30.07
0.0				
82096	0.1600	0.1500	...	-306.57
0.0				
90139	0.1550	0.1450	...	-82.82
0.0				

	LP_GrossPrincipalLoss	LP_NetPrincipalLoss	\
35423	0.00	0.00	
7985	0.00	0.00	
86362	0.00	0.00	
78863	0.00	0.00	
48804	0.00	0.00	
8576	0.00	0.00	
34564	0.00	0.00	
112712	1894.66	1894.66	
82096	0.00	0.00	
90139	0.00	0.00	

	LP_NonPrincipalRecoverypayments	PercentFunded	Recommendations
\			
35423	0.0	1.0	0
7985	0.0	1.0	2
86362	0.0	1.0	0
78863	0.0	1.0	0
48804	0.0	1.0	0
8576	0.0	1.0	0
34564	0.0	1.0	0
112712	0.0	1.0	0
82096	0.0	1.0	1
90139	0.0	1.0	0

	InvestmentFromFriendsCount	InvestmentFromFriendsAmount
Investors		
35423	0	0.0
241		
7985	0	0.0
204		

86362	0	0.0
136		
78863	0	0.0
50		
48804	0	0.0
82		
8576	0	0.0
1		
34564	0	0.0
282		
112712	0	0.0
27		
82096	0	0.0
448		
90139	0	0.0
118		

[10 rows x 81 columns]

```
loan_df = df[['BorrowerAPR',
               'ProsperScore',
               'LoanOriginalAmount',
               'EmploymentStatus',
               'StatedMonthlyIncome',
               'CreditScoreRangeLower',
               'CreditScoreRangeUpper',
               'IncomeRange',
               'LoanStatus',
               'Term']]
```

```
loan_df = loan_df.dropna()
```

```
loan_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 84853 entries, 1 to 113936
```

```
Data columns (total 10 columns):
```

#	Column	Non-Null Count	Dtype
0	BorrowerAPR	84853 non-null	float64
1	ProsperScore	84853 non-null	float64
2	LoanOriginalAmount	84853 non-null	int64
3	EmploymentStatus	84853 non-null	object
4	StatedMonthlyIncome	84853 non-null	float64
5	CreditScoreRangeLower	84853 non-null	float64
6	CreditScoreRangeUpper	84853 non-null	float64
7	IncomeRange	84853 non-null	object
8	LoanStatus	84853 non-null	object
9	Term	84853 non-null	int64

```
dtypes: float64(5), int64(2), object(3)
memory usage: 7.1+ MB
```

```
loan_df.describe()
```

	BorrowerAPR	ProsperScore	LoanOriginalAmount
StatedMonthlyIncome \			
count	84853.000000	84853.000000	84853.000000
8.485300e+04			
mean	0.226658	5.950067	9083.440515
5.931175e+03			
std	0.079916	2.376501	6287.860058
8.239944e+03			
min	0.045830	1.000000	1000.000000
0.000000e+00			
25%	0.163280	4.000000	4000.000000
3.434000e+03			
50%	0.219450	6.000000	7500.000000
5.000000e+03			
75%	0.292540	8.000000	13500.000000
7.083333e+03			
max	0.423950	11.000000	35000.000000
1.750003e+06			

	CreditScoreRangeLower	CreditScoreRangeUpper	Term
count	84853.000000	84853.000000	84853.000000
mean	699.390240	718.390240	42.486135
std	47.095937	47.095937	11.640346
min	600.000000	619.000000	12.000000
25%	660.000000	679.000000	36.000000
50%	700.000000	719.000000	36.000000
75%	720.000000	739.000000	60.000000
max	880.000000	899.000000	60.000000

```
loan_df['AverageCreditScore'] = (
    loan_df['CreditScoreRangeLower'] +
    loan_df['CreditScoreRangeUpper']
) / 2
```

What is the structure of your dataset?

There are 81 columns with 113937 entries.

What is/are the main feature(s) of interest in your dataset?

Analysing the main causes of default on credit.

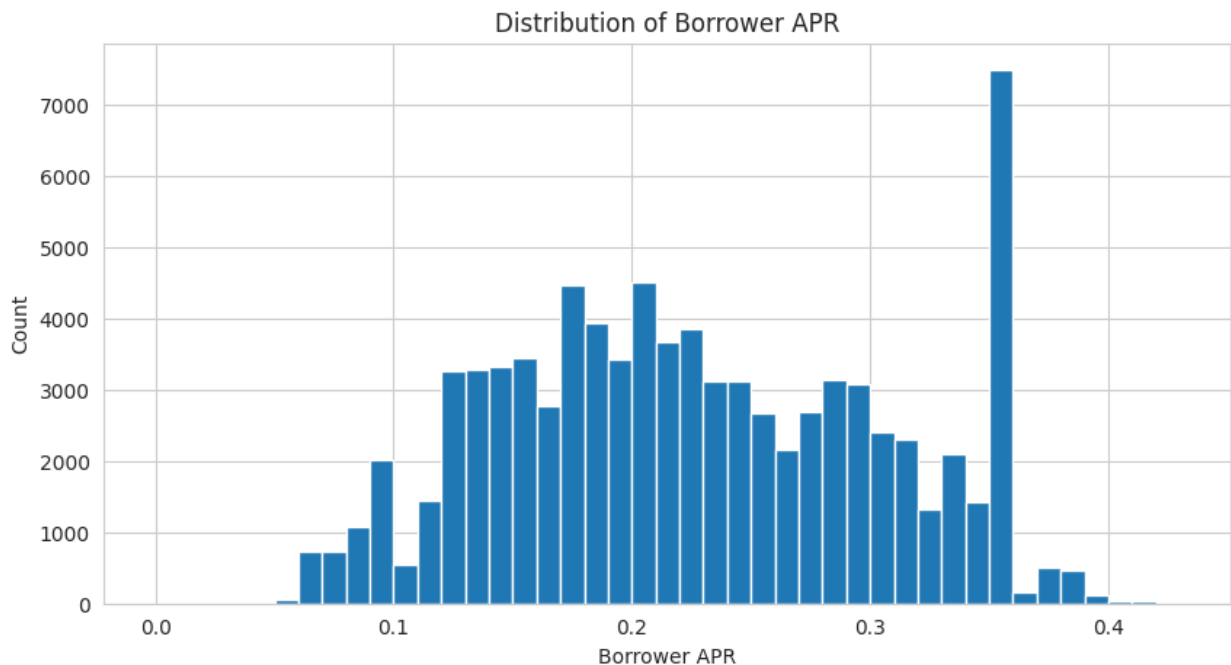
What features in the dataset do you think will help support your investigation into your feature(s) of interest?

Borrowers' prosper rating, the loan amount and employment status may play a major role in defaulting loan.

Univariate Exploration

Distribution of Borrower APR

```
plt.figure(figsize=[10,5])  
  
bins = np.arange(0, loan_df['BorrowerAPR'].max()+0.01, 0.01)  
  
plt.hist(data=loan_df, x='BorrowerAPR', bins=bins)  
  
plt.xlabel('Borrower APR')  
plt.ylabel('Count')  
plt.title('Distribution of Borrower APR');
```

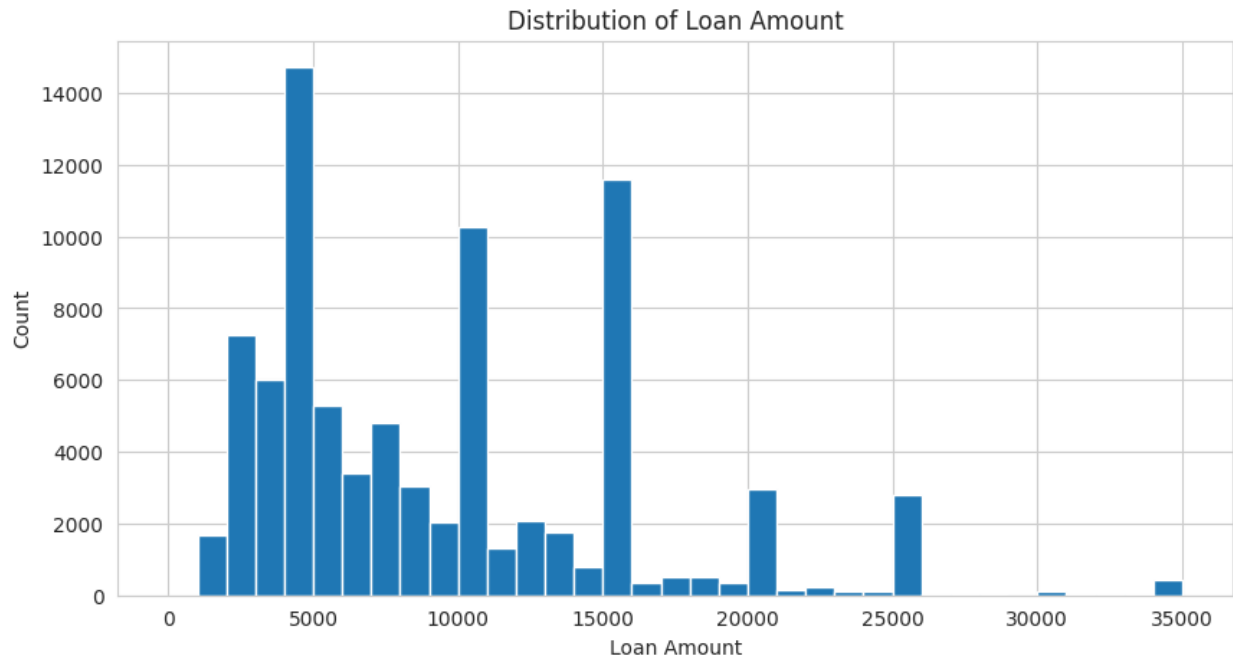


Most borrowers have APR between 10% and 25%

Distribution of Loan Amount

```
plt.figure(figsize=[10,5])  
  
bins = np.arange(0, loan_df['LoanOriginalAmount'].max()+1000, 1000)  
  
plt.hist(data=loan_df, x='LoanOriginalAmount', bins=bins)
```

```
plt.xlabel('Loan Amount')
plt.ylabel('Count')
plt.title('Distribution of Loan Amount');
```



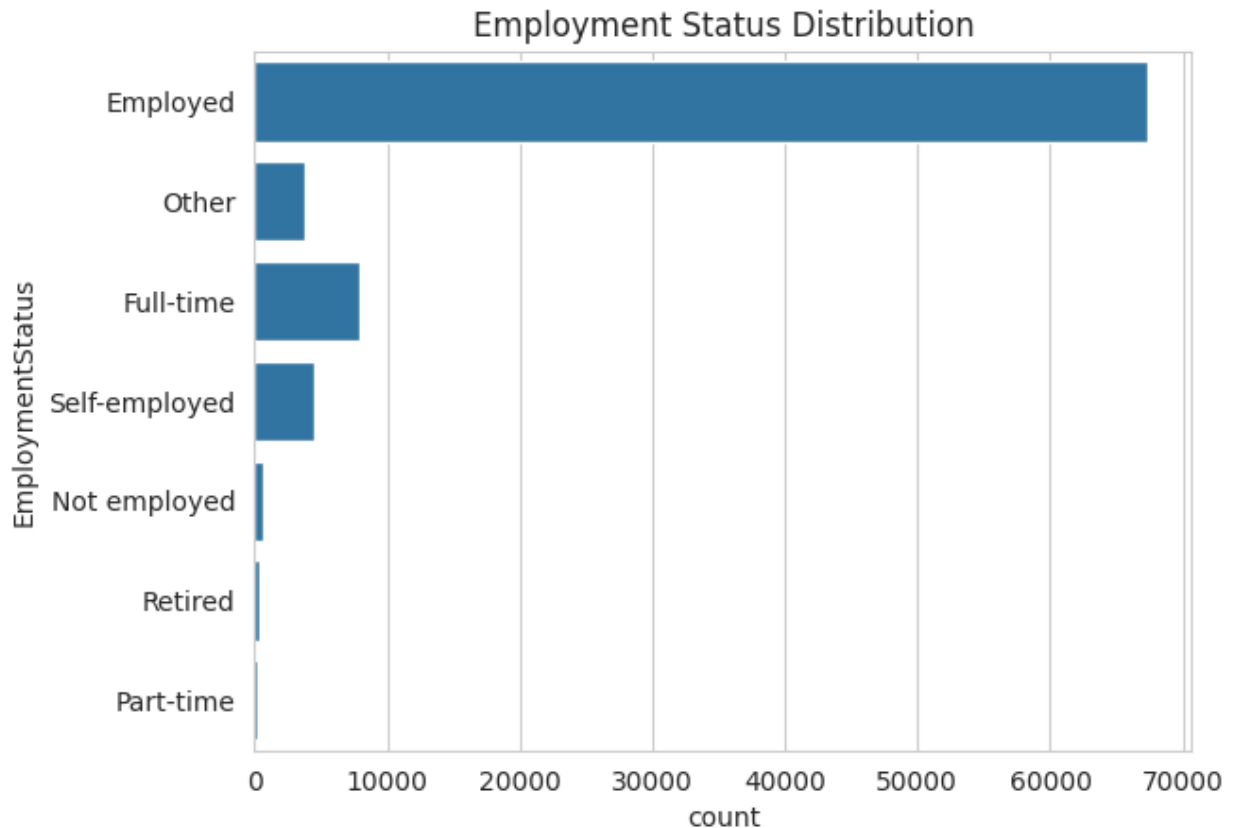
Distribution is skewed to the right.

Employment Status

```
base_color = sns.color_palette()[0]

sns.countplot(data=loan_df,
              y='EmploymentStatus',
              color=base_color)

plt.title('Employment Status Distribution');
```



Most of the borrowers are employed.

Discuss the distribution(s) of your variable(s) of interest. Were there any unusual points? Did you need to perform any transformations?

Majority of borrowers are employed, most borrowers have APR between 10% and 25% and distribution of monthly income is skewed to the right.

Of the features you investigated, were there any unusual distributions? Did you perform any operations on the data to tidy, adjust, or change the form of the data? If so, why did you do this?

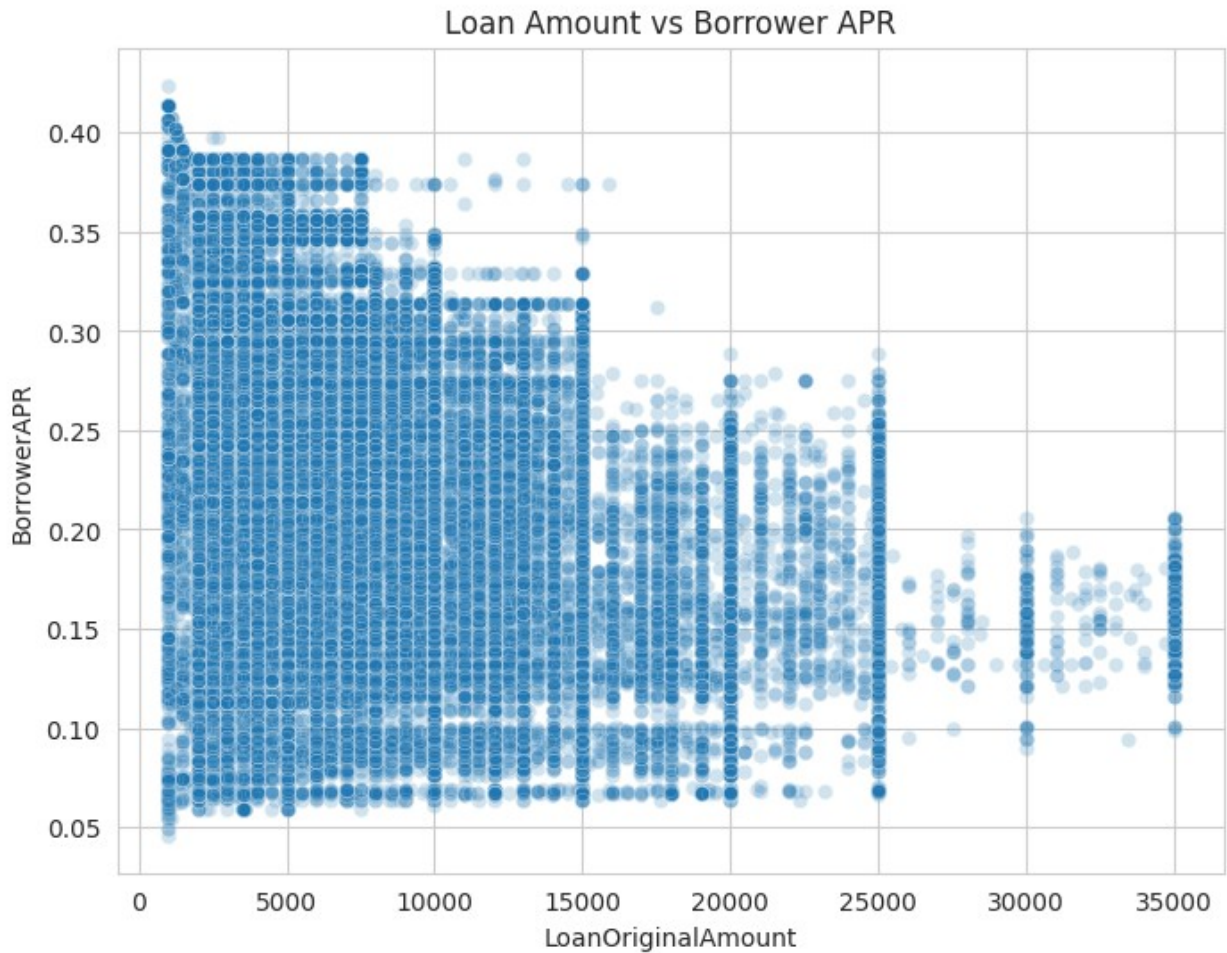
Most borrowers have APR between 10% and 25% and most of the borrowers are employed.

Bivariate Exploration

Loan Amount Vs Borrower APR

```
plt.figure(figsize=[8,6])  
sns.scatterplot(data=loan_df,  
                x='LoanOriginalAmount',
```

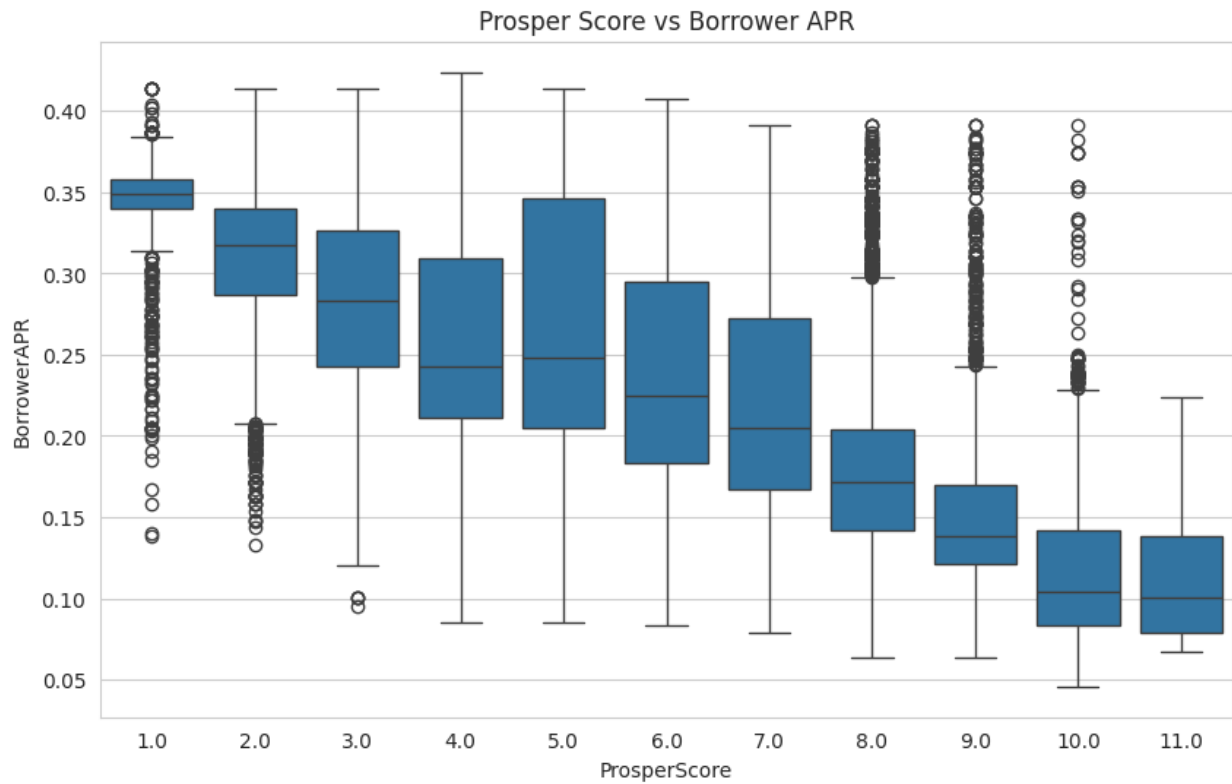
```
y='BorrowerAPR',  
alpha=0.2)  
  
plt.title('Loan Amount vs Borrower APR');
```



Higher loan amounts generally have lower APR.

Prosper Score vs Borrower APR

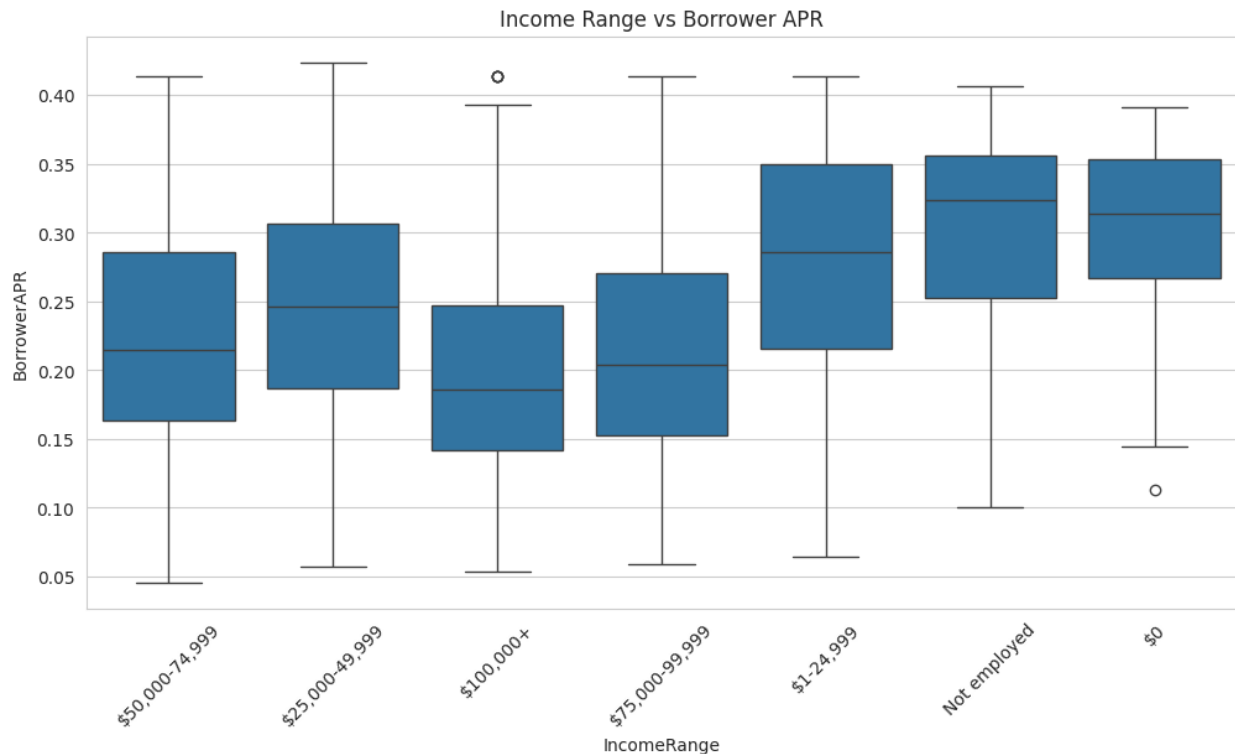
```
plt.figure(figsize=[10,6])  
  
sns.boxplot(data=loan_df,  
            x='ProsperScore',  
            y='BorrowerAPR')  
  
plt.title('Prosper Score vs Borrower APR');
```



Higher Prosper Score leads to lower APR.

Income Range Vs Borrower APR

```
plt.figure(figsize=[12,6])  
sns.boxplot(data=loan_df,  
            x='IncomeRange',  
            y='BorrowerAPR')  
  
plt.xticks(rotation=45)  
plt.title('Income Range vs Borrower APR');
```



Higher-income borrowers tend to receive lower APR.

Talk about some of the relationships you observed in this part of the investigation. How did the feature(s) of interest vary with other features in the dataset?

Higher loan amounts generally have lower APR. Higher Prosper Score leads to lower APR.

Did you observe any interesting relationships between the other features (not the main feature(s) of interest)?

Higher-income borrowers tend to receive lower APR.

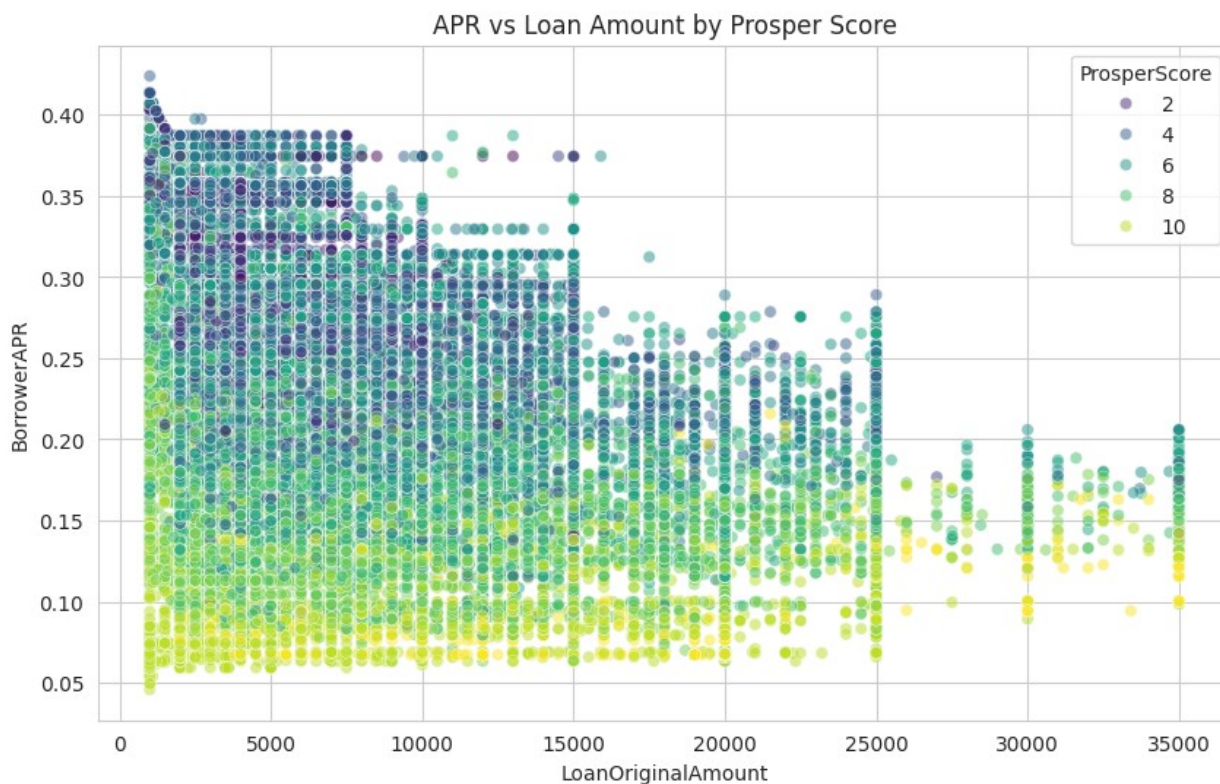
Multivariate Exploration

APR vs Loan Amount by Prosper Score

```
plt.figure(figsize=[10,6])
sns.scatterplot(data=loan_df,
                x='LoanOriginalAmount',
                y='BorrowerAPR',
                hue='ProsperScore',
                palette='viridis',
                alpha=0.5)
```



```
plt.title('APR vs Loan Amount by Prosper Score')
Text(0.5, 1.0, 'APR vs Loan Amount by Prosper Score')
```



Borrowers with high Prosper Scores obtain lower APR even for large loans.

Low-score borrowers experience high APR

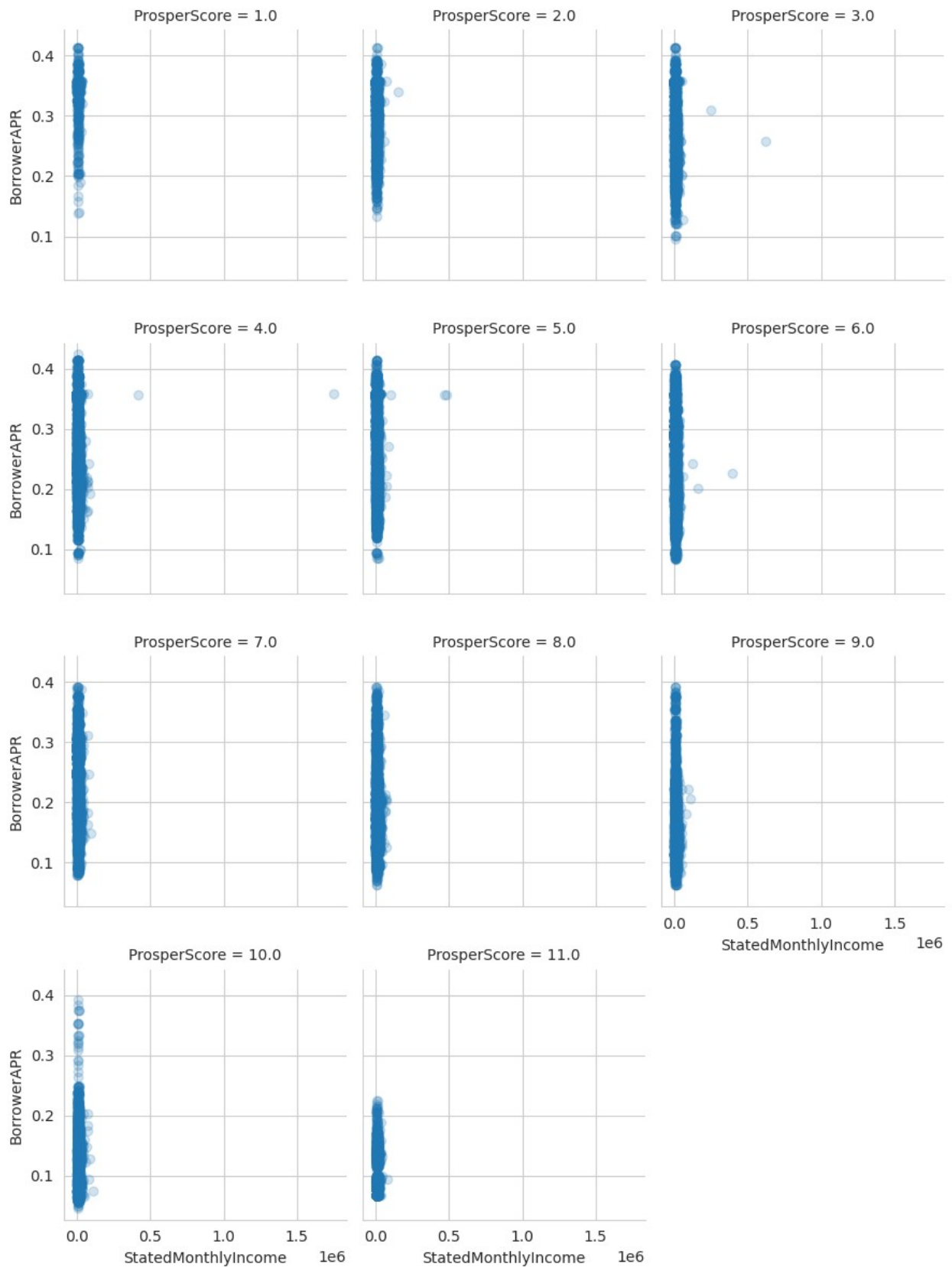
Monthly Income vs APR by Prosper Score

```
g = sns.FacetGrid(data=loan_df,
                  col='ProsperScore',
                  col_wrap=3)

g.map(plt.scatter,
      'StatedMonthlyIncome',
      'BorrowerAPR',
      alpha=0.2)

g.fig.suptitle('Income vs APR by Prosper Score',
              y=1.02);
```

Income vs APR by Prosper Score



Prosper Score strongly impacts loan pricing.

Talk about some of the relationships you observed in this part of the investigation. Were there features that strengthened each other in terms of looking at your feature(s) of interest?

Borrower APR decreases as Prosper Score increases. Employment status and income influence loan approval and pricing. Credit score is strongly associated with lower borrower APR.

Were there any interesting or surprising interactions between features?

Loan amount has a negative relationship with APR.